Sagi **Shaier**

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Research Interests

Artificial intelligence, knowledge graphs, commonsense reasoning, zero-shot & a few-shot learning, multimodal & temporal relations

Languages and Tools: Python, PyTorch, TensorFlow, Scikit-learn, Keras, Pandas, NumPy, SQL, AWS, BigQuery

Education

University of Colorado Boulder *Boulder, Colorado*

PH.D. IN COMPUTER SCIENCE *Aug. 2020 - Present*

Kennesaw State University *Kennesaw, Georgia*

BACHELOR'S OF SCIENCE *Aug. 2014 - Dec. 2018*

• Major in Computational and Applied Mathematics, Statistics minor, Pre-Med track. 3.96 GPA

Professional Experience

Quantum Metric

Data Scientist - Research Team *Dec. 2019 - May. 2020*

- Spearheaded strategic projects such as predictive behavioral analytics, predictive churn risk score, loyalty score, & more
- Researched, constructed, and implemented various algorithms to predict customers' psychological nature across diverse industries such as airlines, retail, & banking
- Integrated Google's BigQuery, AutoML, & servers to optimized queries and analyses on big data while utilizing numerous supervised & unsupervised algorithms such as XGBoost, K-Means, Regressions, DBSCAN, PCA, Random Forest, & Neural Networks

Kepler Computing

Machine Learning Research Engineer (Contract) *July. 2019 - Oct. 2019*

- Identified key computation characteristics of AI workloads in areas such as self-driving, vision, speech, and AR/VR
- Worked with the chief computer architect and HPC engineers to co-design groundbreaking AI ASIC
- Studied cutting edge AI workloads and tracked innovations happening in AI

Welocalize

Machine Learning Intern *May. 2019 - Aug. 2019*

- Built regression models to predict project duration times and to detect outliers
- Gathered, engineered, and visualized various data types
- Identified bottlenecks within the organization using clustering analysis, Seaborn correlations, dimensionality reduction, and H2O.AI RF

Hack Oregon

Data Scientist (Disaster Team) *Feb. 2019 - Oct. 2019*

- Mapped the distribution of likely casualties in the 9.0 Cascadia Earthquake in order to better understand medical response needs

Publications

2019 **Shaier**, Burke, "A Mathematical Model for the Effect of Domestic Animals on Human African Trypanosomiasis (Sleeping Sickness)"

Research Experience & Projects

Human African Trypanosomiasis

Kennesaw State University

Aug 2015 - Dec. 2018

- Pioneered a nine-dimension disease propagation matrix and discovered two factors that offered strong evidence for the eradication of the disease

Waterbender

Water drops detection and removal algorithm

github/Waterbender

Sep. 2019

- Developed a model to enhance the performance of image-based driving assistance systems in rainy weather

DNet

Modified Yolo V3 model to detect objects and their distance from the camera

Video

Jun. 2019

- A real-time detection model utilizing only a single, simple camera

Military

Israel Defense Forces

Israel

Aug 2010 - Aug. 2013

- Prevented further escalations in terrorism throughout the Middle East

Awards & Presentations

Awards

Summer 2021 **Dean's Summer Research Fellowship** University of Colorado Boulder

Fall 2020 **Early Career Professional Development Fellowships** University of Colorado Boulder

Fall 2020 **Departmental Fellowship** University of Colorado Boulder

Fall 2017, Spring 2018, Fall 2018 **National Science Foundation Scholarship** Kennesaw State University

Summer 2018 **Society for Mathematical Biology Award** Kennesaw State University

Summer 2018 **Birla Carbon Scholarship** Kennesaw State University

Spring 2018 **Undergraduate Research Award** Kennesaw State University

Presentations

Jul. 2018	A Mathematical Model for the Effect of Domestic Animals on the Basic Reproduction Number of Human African Trypanosomiasis (Sleeping Sickness)	Annual Meeting of the Society for Mathematical Biology in Sydney, Australia
Apr. 2018	A Mathematical Model for the Effect of Domestic Animals on the Basic Reproduction Number of Human African Trypanosomiasis (Sleeping Sickness)	KSU Symposium of Student Scholars
Jul. 2017	A Mathematical Model for the Effect of Domestic Animals on Human African Trypanosomiasis (Sleeping Sickness)	Annual Meeting of the Society for Mathematical Biology in Utah
Apr. 2017	A Mathematical Model for the Effect of Domestic Animals on Human African Trypanosomiasis (Sleeping Sickness)	KSU Symposium of Student Scholars
Mar. 2017	A Mathematical Model for the Effect of Domestic Animals on Human African Trypanosomiasis (Sleeping Sickness)	Southern Regional Honors Conference
Apr. 2016	A Mathematical Model of African Sleeping Sickness	KSU Symposium of Student Scholars